

Why the two TITE-CRM simulators disagree, and which difference accounts for it

The R and Python implementations produce substantially different MTD selection rates. This note traces that gap to a single design choice, reproduces the published result by adopting it, and lists what still has to be agreed before any comparison of accuracy is meaningful.

SIMULATIONS **1000** PER CELL ACUTE TARGET **0.20** SUBACUTE TARGET **0.33**

CONFIGURATION **NO EWOC, NO BURN-IN**

1 What this note is

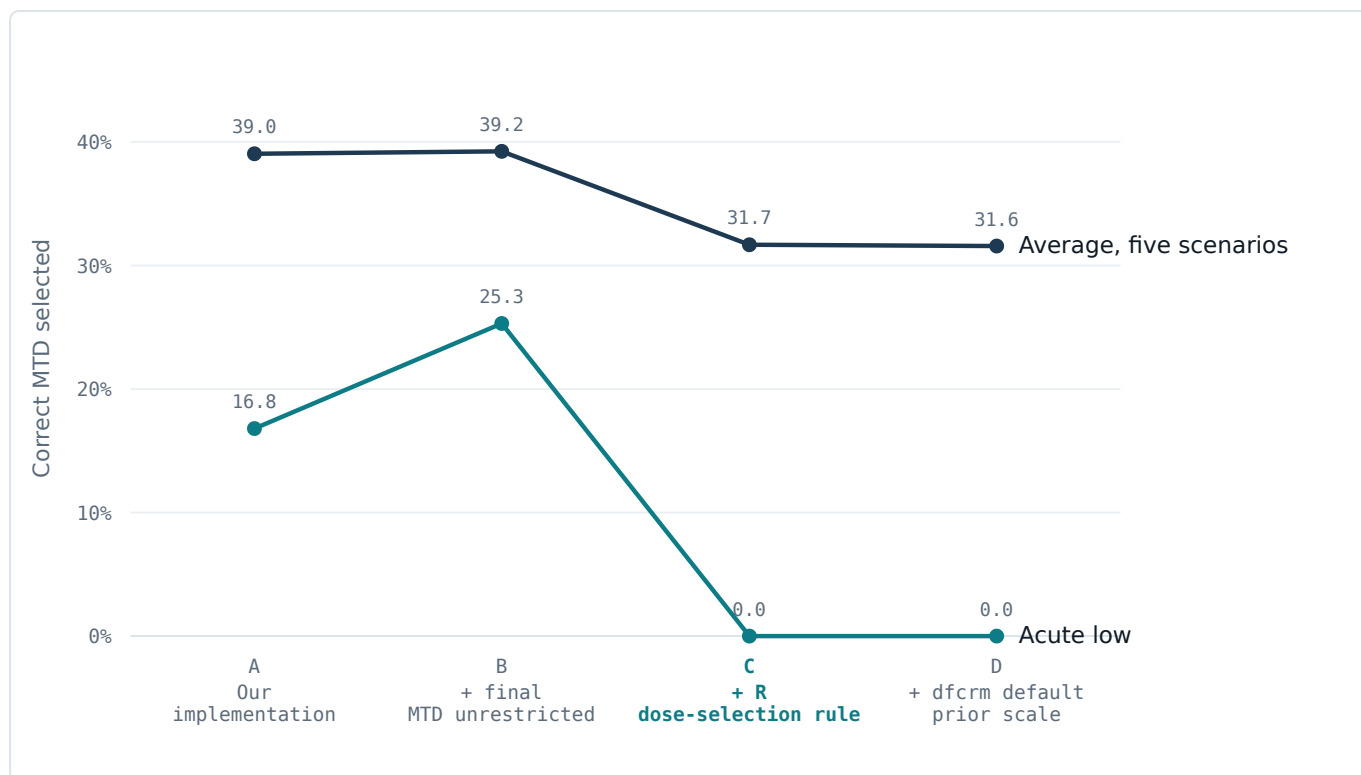
Both groups have simulated what was meant to be the same design, and reached different conclusions about how often the correct MTD is chosen. Before either set of numbers can inform the amendment, we need to know whether the difference comes from the code or from the design.

It comes from the design. The two programs implement genuinely different rules, and one of those rules accounts for almost all of the observed gap. Nothing below is a claim that either implementation is wrong; the point is that they are not yet answering the same question.

The method throughout is the same: start from our configuration, switch one thing at a time towards the R implementation, and measure what changes. Code and outputs are in the repository (`alignment_experiment.py`, `reproduction_check.py`).

2 The dose-selection rule accounts for the gap

Four variants, each adding one difference to the previous one.



Accuracy as each difference is adopted. Removing the restriction to tried doses (A → B) moves the Acute low scenario by 8.5 points and the average barely at all. Adopting the R dose-selection rule (B → C) drops Acute low by 25.3 points, to zero, and takes the average down with it. The prior scale (C → D) changes almost nothing.

CORRECT MTD SELECTION, SWITCHING ONE DIFFERENCE AT A TIME (%)							
VARIANT	LOW	MIDDLE	HIGH	STEEP	SHALLOW	AVERAGE	TOO HIGH
A · Our implementation	16.8	42.9	64.9	45.4	25.2	39.0	17.3
B · + final MTD unrestricted	25.3	44.7	59.5	35.9	30.8	39.2	21.3
C · + R dose-selection rule	0.0	20.1	63.6	64.4	10.3	31.7	8.4
D · + dfcrm default prior scale	0.0	21.0	64.5	62.3	10.1	31.6	8.7

Why the rule excludes the top dose

The acute skeleton is 0.004, 0.021, 0.066, 0.150, 0.266 against a target of 0.20. **The prior at L4 already exceeds the target.** Under the rule “highest dose whose posterior mean is at or below target”, L4 therefore sits outside the admissible set from the very first decision, and the data would have to pull that estimate below 0.20 before it can ever be chosen. With one DLT at L1 raising the whole curve, that does not happen within 30 patients.

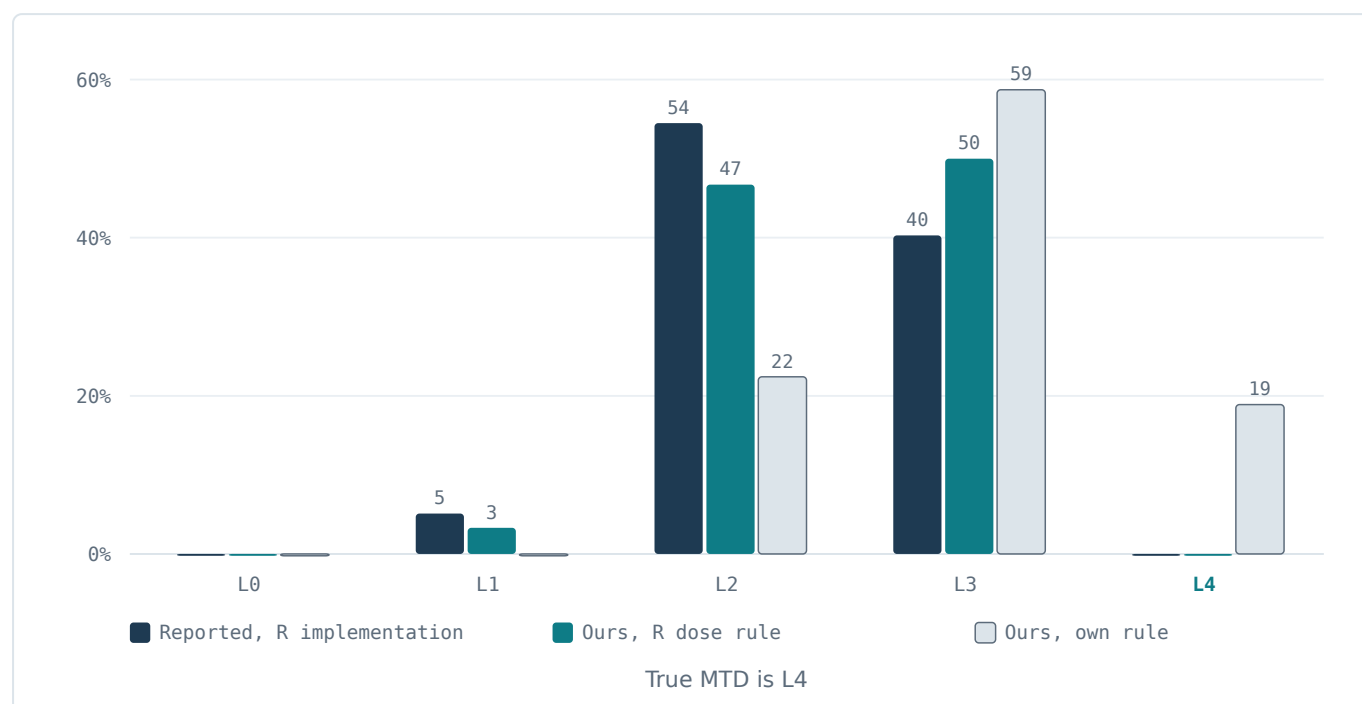
We checked whether the subacute endpoint contributes by relaxing its target to 0.99. The result is unchanged at 0.0%, so this is the acute rule alone.

NOTE ON THE INTERACTION WITH THE CONTESTED DLT

This rule and the single contested toxicity at L1 compound each other. The rule makes the top dose inadmissible until the estimate falls; the DLT keeps the estimate up. Either on its own would be recoverable. Together they close off the top dose for the whole trial.

3 Reproduction check

If the rule really is the explanation, then adopting it should reproduce the published distribution, not merely move in its direction. It does.



Acute low, where the true MTD is the top dose. Our own rule selects L4 in 18.9% of trials. With the R rule that becomes 0.0%, matching the reported 0.0%, and the rest of the distribution moves close to the reported shape as well. The remaining differences are consistent with the other design differences listed in section 5.

ACUTE LOW: FINAL DOSE SELECTED (%)					
SERIES	L0	L1	L2	L3	L4
Reported, R implementation	0.0	5.1	54.5	40.3	0.0
Ours, own rule	0.0	0.0	22.4	58.7	18.9
Ours, R dose rule	0.0	3.3	46.7	50.0	0.0

4 Errors on our side

Three things we got wrong, listed before anything about the R code, because two of them affect results we have already circulated.

1 · OUR SUBACUTE PRIOR WAS WRONG

The R code uses the exact skeletons from the 8 July message, including a subacute prior of **0.010** at L0. Ours read 0.012. Ours was the deviating value and has been corrected. The effect is negligible — trials start at L2 and rarely visit L0 — but earlier outputs were generated with the wrong figure.

2 · WITH EWOC OFF, OUR SUBACUTE ENDPOINT DOES NOTHING

In our dose-selection function the subacute overdose probability is computed and then never used when EWOC is disabled. The subacute endpoint therefore has no influence on dose selection at all in that configuration — which is the configuration every recent analysis of ours has run under. In the R code both endpoints always bind. This is a design question, not only a coding one: should the subacute endpoint constrain dose selection when EWOC is off?

3 · OUR 6+3 COMPARISON WAS NOT LIKE FOR LIKE

The R example runs the rule-based design with a sample size of 50 and an expansion concept (`target.patients.at.mtd = 12`) that our implementation does not have; we gave ours 30 patients. Our figure of 2.8% correct selection against the reported 32.3% should therefore **not** be read as contradicting the R result.

5 Differences we found but have not quantified

Each of these is a real difference in what is being simulated. None has been isolated the way the dose rule was, so their contributions are unknown.

Surgery is certain, and a delay can become a DLT

In the R model every patient is operated: the surgery time matrix is initialised at `operation.Time` for everyone. Our model draws surgery from a Bernoulli with probability 0.80, and patients without surgery contribute no subacute information at all.

More consequentially, a surgery delayed beyond `Max.Surgery.Time` is **recorded as an acute DLT**, reached through a non-DLT toxicity occurring with probability 0.15 that we do not model at all. This means the realised acute toxicity in the R simulation is higher than the specified `P.early`, and so the true MTD derived from `P.early` is strictly not the true MTD of the data-generating process. That affects the interpretation of every accuracy figure on both sides, and is the first thing we would want to confirm.

The run-in phase does not escalate

In the R code the dose is fixed at the starting level throughout the first phase and the loop always runs to six patients, because the stopping condition cannot trigger before patient six. Ours escalates one level per cohort until the first observed acute DLT, as described in July. The two “burn-in on/off” comparisons therefore do not measure the same thing and should not be placed side by side.

Accrual

The R code enrolls a whole cohort at one shared entry time, with the next cohort a fixed interval later; ours uses a Poisson process with individual arrivals. Within-cohort TITE weights are therefore identical on the R side and not on ours. The time unit of `time_para` is not stated anywhere we could find, so we have not tried to compare the rates.

Smaller items

- De-escalation is unrestricted in the R code; ours is capped at one level per cohort.
- The final MTD may be one level above the *last treated* dose — not the highest ever tried — so it can be a dose nobody received, and can also exclude a higher dose that was tried earlier.
- A Gaussian copula links the two endpoints on the R side (set to independent in the example); we do not model any dependence.
- The prior scale is the `dfcrm` default, $\sqrt{1.34}$, against our 1.0. Same empiric model; measured effect here is negligible.
- Toxicity is reported as a proportion on the R side and as a count on ours. These agree once converted.

6 Two things that look like defects

Offered for checking rather than asserted — both are the kind of thing that survives testing precisely because it is invisible in the configurations run so far.

UNDEFINED ARGUMENT NAMES

`real.data.early` and `real.data.late` are passed to the trial function, but the parameters are named `real.dlt.early` and `real.dlt.late`, and the names passed do not exist anywhere. This does not error today because R evaluates arguments lazily and the trial function never uses them.

THE RUN-IN OVERWRITES THE FIXED HISTORY'S DOSE

The first phase assigns the starting dose to patients one through six regardless of what was supplied in `real.dose`. For the MERGE configuration the two coincide, so nothing goes wrong in practice. The recorded outcomes of those patients are used correctly.

7 Questions we would like to settle

In order of measured or expected effect.

1. **The dose-selection rule.** With the current skeleton, “highest dose at or below target” excludes L4 from the outset because the prior there is already 0.266. Is that intended? This drives most of the difference and affects the whole allocation, not only the final choice.
2. **Does a surgery delay beyond the maximum count as an acute DLT?** If so, the realised acute toxicity exceeds `P.early` and the reference MTD needs rethinking.
3. **Should the subacute endpoint constrain dose selection** when EWOC is off? It does in the R code and does not in ours.
4. **Should every patient be operated,** or should surgery be probabilistic as in our model?
5. **May the final MTD be a dose nobody received?**
6. **Is the prior scale a choice or the package default?**
7. **Should the first phase escalate** until the first DLT, or treat six patients at the starting dose?
8. **Which accrual structure, time unit and 6+3 settings** should the amendment use?

8 Conclusion

WHERE THIS LEAVES THE COMPARISON

The two simulators are not two implementations of one design. They differ on the dose rule, the role of the subacute endpoint, whether surgery is certain, what counts as an acute DLT, the accrual structure, the run-in, the prior scale and the 6+3 settings.

Comparing accuracy percentages between them is therefore not yet meaningful, and we would not use either set of numbers to argue the other is wrong.

What has been established is where the difference comes from, and that it can be reproduced on demand. The natural next step is to agree which rules represent the intended MERGE design, align both implementations on those, and re-run. Most of the alignment is small: the dose rule, the prior scale and the subacute constraint are each a few lines.

A patient-level generating file for a single trial is already prepared on our side, and now that the data structures on both sides are understood it can be mapped across directly. Running one identical trial through both programs would confirm the alignment before any larger comparison is attempted.

Generated from `alignment_experiment.py` and `reproduction_check.py`, 1000 simulations per cell. TITE-CRM without EWOC and without burn-in, starting at L2, 30 patients including six with fixed history, cohort size 3, one acute DLT in the sixth of those. Reported R figures are taken from the circulated slides for the corresponding configuration.

This document contains aggregated simulation output only. No patient data is included.